TEXORA · LECTURE SHEET

Yarn Singeing

Every staple yarn is covered with short hairs — fibre ends sticking out of its surface. Singeing burns those hairs off in a flame the yarn passes through in a fraction of a second, leaving the yarn smooth, lustrous and clean-printing. This sheet covers the process, the machine, its settings and its risks.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 9 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. Purpose of singeing

Singeing (also called gassing, because the classic machine burns gas) is the controlled burning-off of fibres protruding from a yarn (or fabric) surface. The yarn body itself must not be scorched: the process works because a thin protruding hair heats and ignites almost instantly in the flame, while the compact yarn body, moving fast, passes through before it can take damage — provided speed, flame and tension are correctly set.

Singeing is carried out to:

- **Remove protruding hairs**, giving a smooth, clean yarn surface with a clearer definition of twist and structure.
- **Improve lustre.** Hairs scatter light in all directions and make a yarn look dull and fuzzy; removing them lets the yarn surface reflect light evenly, so lustre rises.
- **Improve later processes.** A smooth yarn passes more easily through heddles, guides and needles, sheds less lint and fly, and gives sharper printed patterns and cleaner dyed shades — loose hairs dye and print as a fuzzy halo around the design.
- **Improve fabric appearance and handle.** Fabrics from singed yarn look crisper, pill less readily (fewer loose ends are available to form pills) and show weave or stitch structure clearly — the reason singed (gassed) yarns are specified for quality shirting, poplin, sewing threads and fine knitwear.

Singeing is applied mainly to cotton and other cellulosic staple yarns and to some blends. It is not applied to filament yarns, which have no staple hairs to remove.

2. The gas singeing machine

2.1 Yarn path

The machine is essentially a precision winder whose yarn path crosses one or more gas flames between the supply package and the take-up package:

1. **Creel / supply package.** The unsinged yarn package is mounted so the yarn unwinds freely without tension spikes.
2. **Tensioner and cleaner.** A tension device keeps the yarn taut and straight — a slack yarn sags into the flame unevenly — and a guide aligns it exactly with the burner.
3. **Burner.** The yarn passes through or immediately over a row of small, even gas flames (or a slotted burner giving a sheet of flame). Gas and air are mixed to give a hot, clean, blue, non-smoking flame; a yellow, sooty flame would deposit carbon on the yarn.
4. **Guides after the flame** carry the singed yarn clear of the burner; on some machines the yarn crosses a second burner for a second passage within the same pass.
5. **Winder.** Delivery rollers set the yarn speed and the yarn is wound onto a cone or cheese, ready for weaving, knitting or further processing.

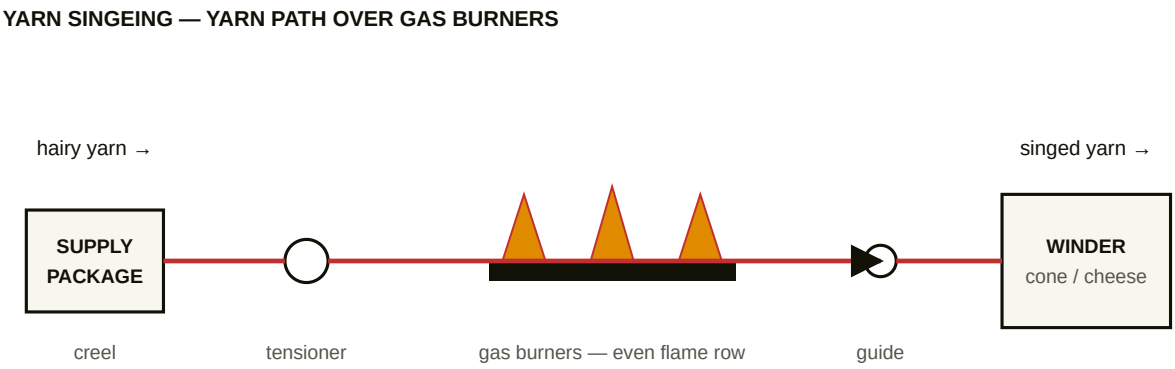


Figure 1 — Gas singeing machine. The taut yarn runs from the creel through a tensioner, across an even row of gas flames, and through a guide to the winder.

2.2 Number of passages and speed

One passage through one flame rarely removes every hair, because hairs lying flat against the yarn or shielded on the far side escape. Machines therefore give the yarn **more than one flame contact**: either two burners in one pass, or the yarn is singed, rewound and singed again. Fine, high-quality yarns commonly receive two passages; more passages give diminishing returns and rising damage risk, so the number is set by the hairiness specification, not by habit.

Yarn speed is the second control. Singeing speeds are high — typically several hundred metres per minute, of the order of 400–800 m/min as typical working values — and speed and flame intensity

must be matched: too slow (or too fierce a flame) scorches and weakens the yarn body; too fast (or too weak a flame) leaves hairs behind. Because different positions on the machine can drift, flame evenness across all burners is checked routinely: one weak burner produces one faulty package among good ones.

3. Singeing of single versus doubled yarn

- **Single yarn singeing.** The singles yarn is singed before doubling. Each strand enters plying already smooth, and the plied yarn is very clean — but singeing singles is delicate work: the singles yarn is at its weakest and most hairy, and flame damage at this stage is locked into every ply.
- **Doubled (plied) yarn singeing.** The yarn is doubled and ply-twisted first, then singed as one strand. The plied yarn is stronger and rounder, tolerates the flame more safely, and singeing after plying also burns off hairs raised by the twisting process itself. Most gassed sewing threads and quality plied yarns are singed in the doubled state.

In both cases the logic is the same: singe as late as the product allows, so that hairs created by later handling are also removed, but never after a process (such as dyeing) that the flame or its heat would disturb.

4. Effects on yarn properties

PROPERTY	EFFECT OF CORRECT SINGEING	NOTE
Hairiness	Greatly reduced	The purpose of the process; verify by hairiness testing
Lustre	Increased	Even reflection from the smooth surface
Count / mass	Slight loss	Burnt hairs are lost mass; allow for it in costing
Strength	Nearly unchanged if correct; falls if overdone	See precautions, Section 5
Evenness of appearance	Improved	Twist and structure show clearly
Dyeing / printing	Cleaner, sharper shades	No fuzzy halo of dyed hairs
Pilling in fabric	Reduced	Fewer loose fibre ends available

WORKED EXAMPLE — SINGEING PRODUCTION (TYPICAL VALUES)

A singeing position runs at 600 m/min on a 30 tex yarn. These are **typical illustrative values**, not a measurement from a specific machine.
Length in 8 h = 600 × 60 × 8 = **288,000 m**.
Mass = 288,000 × 30 / 1,000,000 = **8.64 kg per position per 8 h**.
Hairiness outcome, stated as a typical result: a yarn entering with a hairiness index of about 6 commonly leaves at about 2–3 after two passages — roughly a halving or better. Treat such figures as typical ranges to be confirmed by testing the actual yarn, never as guaranteed values.

5. Precautions

- **Over-singeing weakens yarn.** If the yarn body itself is heated too long, cellulose degrades, the yarn yellows and tensile strength falls — a singed yarn that breaks in weaving was usually too slow through too strong a flame. Strength-test singed yarn against its unsinged parent as a routine check.
- **Flame evenness.** Gas pressure, air mixing and burner cleanliness must give identical flames at every position; check for blocked burner holes and soot, which mark and weaken the yarn locally.
- **Tension and alignment.** The yarn must run taut and exactly through the flame centre at every position; a wandering yarn is singed on one side only and shows a hairy stripe in fabric.
- **Moisture and cleanliness.** Yarn should be in a normal, dry, conditioned state; wet or dirty yarn singes unevenly. Lint and burnt fibre accumulate around burners and must be cleaned — they are both a quality fault and a fire risk.
- **Safety.** Open flames, gas supply and fine airborne fibre demand extraction, guarded burners, automatic flame-failure shut-off and strict housekeeping.

6. Alternatives and related processes

The best-known alternative works one stage later, on fabric: **biopolishing**. Cellulase enzymes hydrolyse the protruding cellulose microfibrils on a cotton fabric surface, so the hairs are digested away rather than burnt, giving a smooth, soft, low-pilling fabric. It acts on fabric (and garments), not on yarn, adds a wet process with weight and strength loss to control, and works only on cellulosic fibres — it complements singeing rather than replacing it for yarn. Mechanical hair removal (shearing/cropping) serves the same surface-cleaning idea on fabric. For yarn itself, gas singeing remains the standard method.

Glossary

Singeing / gassing — burning protruding fibres off a yarn or fabric surface with a controlled flame. **Hairiness** — amount of fibre ends and loops protruding from a yarn surface. **Gassed yarn** — yarn that

has been singed. **Burner** — the gas jet or slot producing the singeing flame. **Passage** — one passage of the yarn through a flame. **Biopolishing** — enzyme (cellulase) removal of surface microfibrils from cellulosic fabric. **Fly / lint** — short fibres released into the air or onto surfaces during processing.

Key points

• Singeing burns off protruding hairs without damaging the yarn body — hairs ignite instantly; the fast-moving compact body does not.
• Gains: smoothness, lustre, cleaner dyeing and printing, less lint, less pilling, clearer fabric structure.
• Machine path: creel → tensioner → even gas flame(s) → guide → winder; typically two flame contacts, at several hundred metres per minute (typical values).
• Doubled yarn is usually singed after plying — stronger, rounder, and post-twisting hairs are removed too.
• Over-singeing yellows and weakens yarn; flame evenness, tension and alignment decide quality.
• Biopolishing is the fabric-stage enzyme alternative for cellulosics — related in purpose, different in stage and mechanism.

Practice

1. A singeing position runs at 500 m/min on a 20 tex yarn (typical illustrative values). Calculate the length and the mass of yarn singed in 8 h.
2. A singed warp keeps breaking on the loom although the unsinged yarn wove well. Give the most likely singeing cause and the two machine settings you would check first.
3. Explain why two passages singe better than one passage at double flame strength, referring to how hairs lie on the yarn.
4. A mill must choose between singeing the singles yarn and singeing after doubling for a plied sewing thread. State the choice you recommend and give two reasons.

Answer key

1. Length = 500 × 60 × 8 = **240,000 m**. Mass = 240,000 × 20 / 1,000,000 = **4.8 kg per position per 8 h**.
2. Most likely over-singeing: the yarn body has been heat-damaged, losing strength. Check first (1) yarn speed — too slow for the flame — and (2) flame intensity / gas–air setting — too fierce. Confirm by tensile-testing singed against unsinged yarn.
3. Many hairs lie flat against the yarn or are shielded behind it and never enter the hottest part of a single flame, however strong it is; a stronger flame only scorches the yarn body sooner. A second

passage presents the yarn again — hairs lifted or turned by the first passage and winding are caught the second time — so removal improves without extra damage to the body.

4. Singe after doubling. Reasons: the plied yarn is stronger and rounder, so it tolerates the flame with less damage risk; and singeing after plying also removes hairs raised by assembly winding and twisting, giving the cleanest finished thread.
